

## 苏州苏德兰铁路配件有限公司尺寸检测报告

## DIMENSIONAL TEST REPORT

品名 Product Name: 螺纹道钉 Screw Spike

合同编号 Contract NO. PO009475

规格 Size:  $\Phi 25 \times 164$ 

检测日期 Inspection date: 2026.05.15

等级 Grade: 8.8, UIC864-1

发货日期 Date of shipment: 2026.06.02

标记 Mark: PANDROL 26 8.8 GS195-J8 SDL



| 序号<br>No     | 检测项目<br>Test feature            | 技术要求<br>Specification             | 实测数值<br>Actual Size                                        | 结果<br>Result |
|--------------|---------------------------------|-----------------------------------|------------------------------------------------------------|--------------|
| 1            | 对边尺寸<br>opposite side           | 27 +0/-1                          | 26.1~26.4                                                  | OK           |
| 2            | 头部厚度<br>head thickness          | 15.5±0.5                          | 15.3~15.7                                                  | OK           |
| 3            | 头部R<br>radius of head           | R2                                | R2                                                         | OK           |
| 4            | 头部R<br>radius of head           | R2±0.5                            | R2                                                         | OK           |
| 5            | 头部角度<br>Head Angle              | 15°                               | 15°                                                        | OK           |
| 6            | 头部总高<br>head total length       | 20 +1.5/-0                        | 20.5~20.9                                                  | OK           |
| 7            | 圆盘直径<br>head diameter           | $\Phi 40$ +0/-1                   | 39.4~39.7                                                  | OK           |
| 8            | 圆盘直径<br>head diameter           | $\Phi 42$ +3/-1                   | 41.4~41.8                                                  | OK           |
| 9            | 杆部直径<br>shank diameter          | $\Phi 25$ +0.7/-0.5               | 24.8~25.1                                                  | OK           |
| 10           | 白杆长度<br>unthreaded shank length | 60±5                              | 59.5~62.2                                                  | OK           |
| 11           | 杆部总长<br>shank length            | 164±3                             | 162.7~166.3                                                | OK           |
| 12           | 螺纹内径<br>ID                      | $\Phi 19$ +0/-1                   | 18.2~18.5                                                  | OK           |
| 13           | 螺纹外径<br>OD                      | $\Phi 25$ +0/-1                   | 24.2~24.5                                                  | OK           |
| 14           | 5牙距<br>screw pitch              | 60±0.5                            | 59.7~60.2                                                  | OK           |
| 15           | 单牙距<br>single screw pitch       | 12±0.1                            | 11.95~12.05                                                | OK           |
| 16           | 螺纹底部R角<br>Thread R angle        | R0.5 +0.5/-0                      | R1                                                         | OK           |
| 17           | 螺纹底部R角<br>Thread R angle        | R1 +1/-0.5                        | R1                                                         | OK           |
| 18           | 螺纹头部R角<br>Thread R angle        | R0.5 +0.5/-0                      | R1                                                         | OK           |
| 19           | 螺纹头部R角<br>Thread R angle        | R1 +1/-0.5                        | R1                                                         | OK           |
| 20           | 牙高<br>screw height              | 3                                 | 2.9~3.1                                                    | OK           |
| 21           | 螺纹角度<br>Thread angle            | 30°                               | 30°                                                        | OK           |
| 22           | 螺纹角度<br>Thread angle            | 30°                               | 30°                                                        | OK           |
| 23           | 尖部深度<br>Point depth             | ≤8                                | 4.5~6.0                                                    | OK           |
| 24           | 表面处理<br>surface treatment       | 电镀白锌 ≥12 μm<br>White Zinc plating | 盐雾试验200小时无红锈<br>Salt spray test 200 hours without red rust | OK           |
| 数量 Quantity  |                                 |                                   | 12件 (Pieces)                                               |              |
| 检验 Inspector |                                 |                                   | 张全恢 Zhang-QuanHui                                          |              |
| 审核 In Charge |                                 |                                   | 杨黎程 Yang-LiCheng                                           |              |



苏州苏德兰铁路配件有限公司性能检测报告

PERFORMANCE TEST REPORT

品名 Product Name: 螺纹道钉 Screw Spike

规格 Size:  $\Phi 25 \times 164$

等级 Grade: 8.8, UIC864-1

标记 Mark : PANDROL 26 8.8 GS195-J8 SDL

合同编号 Contract NO

P0009475

检测日期 Inspection date: 2026.05.15

发货日期 Date of shipment: 2026.06.02

数量

Quantity: 12(件 Pieces)



| 规格<br>Size | 材料<br>Material | 炉批号<br>Raw material<br>Heat No | 化学成分<br>Chemical composition ( % ) |       |      |      |       | 机械性能<br>Mechanical properties |                             |                          |                 |                |                                          |
|------------|----------------|--------------------------------|------------------------------------|-------|------|------|-------|-------------------------------|-----------------------------|--------------------------|-----------------|----------------|------------------------------------------|
|            |                |                                | C                                  | S     | Si   | Mn   | P     | 屈服强度<br>Yield<br>strength     | 抗拉强度<br>Tensile<br>Strength | 延伸率<br>Elongation<br>δ5" | 冷弯<br>Bend Test | 硬度<br>Hardness | 断面收缩率<br>Rate of<br>reduction<br>in area |
| Φ 25×164   | 45#            | B26020174                      | 0.46                               | 0.004 | 0.26 | 0.62 | 0.023 | ≥660Mpa                       | ≥830Mpa                     | Min12%                   | 15°             | HRC23-34       | ≥52 %                                    |
|            |                |                                |                                    |       |      |      |       | 671                           | 875                         | 16                       | OK              | 27             | 58                                       |
|            |                |                                |                                    |       |      |      |       | 675                           | 887                         | 16                       | OK              | 28             | 57                                       |

检验 Inspector: 张全恢 Zhang-QuanHui

审核 In Charge: 杨黎程 Yang-LiCheng





浙江元立金属制品集团有限公司

皖CG1827

ZHEJIANG YUANLI METAL PRODUCT GROUP CO., LTD  
产品质量证明书  
Product Quality Certificate

浙江. 遂昌. 上江工业园区  
ADD: NO. 181 YUANLI ROAD, MIAOGAO STREET, SUICHANG, LISHUI, ZHEJIANG  
TEL: 18157815951  
FAX: (0578) 8195050  
邮编: 323300  
POST CODE: 323300

|                      |              |                            |        |                          |            |
|----------------------|--------------|----------------------------|--------|--------------------------|------------|
| 订货单位<br>Purchaser    | 上海元立实业有限公司   | Steel 钢号                   | 45#-XZ | 出库单号<br>Deliver NO.      | 260303080  |
| 产品名称<br>Product Name | 热轧无扭控冷盘条     | 总重量(t)<br>Total WEIGHT     | 26.440 | 证书号<br>Certification     | 11083908-2 |
| 执行标准<br>Standard     | GB/T699-2015 | 交货状态<br>Delivery Condition | 热轧     | 签发日期<br>Date of Delivery | 2026-03-03 |

| 序号 | 炉批号<br>Heet NO. | 规格<br>(mm)<br>SIZE | 件数<br>QTY | 钢号<br>Steel<br>Grade | 化学成份 Chemical Composition % |      |      |       |       |       |       |       |       |       |       |       |        |        |        |          | 低倍组织 (级) |    |     | 离淬火端硬<br>度值(HRC) |  |
|----|-----------------|--------------------|-----------|----------------------|-----------------------------|------|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------|--------|--------|----------|----------|----|-----|------------------|--|
|    |                 |                    |           |                      | C                           | Si   | Mn   | P     | S     | Cr    | Ni    | Mo    | Cu    | Al    | Ti    | V     | B      | W      | Zr     | 一般<br>疏松 | 中心<br>疏松 | 偏析 | 9mm | 15mm             |  |
| 1  | B26020173       | Φ26                | 1         | 45#-XZ               | 0.46                        | 0.23 | 0.57 | 0.017 | 0.007 | 0.067 | 0.013 | 0.004 | 0.033 | 0.008 | 0.001 | 0.002 | 0.0001 | 0.0000 | 0.0003 |          |          |    |     |                  |  |
| 2  | B26020174       | Φ26                | 11        | 45#-XZ               | 0.46                        | 0.26 | 0.62 | 0.023 | 0.004 | 0.054 | 0.015 | 0.006 | 0.029 | 0.010 | 0.002 | 0.004 | 0.0004 | 0.0000 | 0.0012 |          |          |    |     |                  |  |
| 3  |                 |                    |           |                      |                             |      |      |       |       |       |       |       |       |       |       |       |        |        |        |          |          |    |     |                  |  |
| 4  |                 |                    |           |                      |                             |      |      |       |       |       |       |       |       |       |       |       |        |        |        |          |          |    |     |                  |  |
| 5  |                 |                    |           |                      |                             |      |      |       |       |       |       |       |       |       |       |       |        |        |        |          |          |    |     |                  |  |
| 6  |                 |                    |           |                      |                             |      |      |       |       |       |       |       |       |       |       |       |        |        |        |          |          |    |     |                  |  |
| 7  |                 |                    |           |                      |                             |      |      |       |       |       |       |       |       |       |       |       |        |        |        |          |          |    |     |                  |  |

| 序号                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 物理性能 Physical Capability |             |          |          |            | 顶锻<br>Upsetting | 显微组织<br>Microstructure |             | 带状<br>组织<br>(级) | 碳化物 (级) |    |    |    | 非金属夹杂物(级) Non Metallic inclusion |    |    |    |     |    |    |    | 备注  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------------|----------|----------|------------|-----------------|------------------------|-------------|-----------------|---------|----|----|----|----------------------------------|----|----|----|-----|----|----|----|-----|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | ReI/Rp0.2<br>(Mpa)       | Rm<br>(Mpa) | A<br>(%) | Z<br>(%) | Akv<br>(J) |                 | 晶粒度<br>(级)             | 脱碳层<br>(mm) |                 | 网状      | 带状 | 液析 | 粗系 | 细系                               | 粗系 | 细系 | 粗系 | 细系  | 粗系 | 细系 | 粗系 | 细系  |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 389                      | 678         | 21.0     | 43       |            | 合格              | 9.0                    | 0.130       |                 |         |    |    |    | 1.0                              |    |    |    |     |    |    |    |     |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 380                      | 679         | 21.0     | 44       |            | 合格              | 9.0                    | 0.130       |                 |         |    |    |    | 1.0                              |    |    |    |     |    |    |    | 1.0 |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                          |             |          |          |            |                 |                        |             |                 |         |    |    |    |                                  |    |    |    |     |    |    |    |     |
| 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                          |             |          |          |            |                 |                        |             |                 |         |    |    |    |                                  |    |    |    |     |    |    |    |     |
| 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                          |             |          |          |            |                 |                        |             |                 |         |    |    |    |                                  |    |    |    |     |    |    |    |     |
| 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                          |             |          |          |            |                 |                        |             |                 |         |    |    |    |                                  |    |    |    |     |    |    |    |     |
| 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                          |             |          |          |            |                 |                        |             |                 |         |    |    |    |                                  |    |    |    |     |    |    |    |     |
| 总计[Total] 12 件                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                          |             |          |          |            |                 |                        |             |                 |         |    |    |    |                                  |    |    |    |     |    |    |    |     |
| 1.材料的表面质量和外形尺寸均合格, 酸浸低倍组织中白点、宏观夹杂等标准规定不允许有的缺陷, 均保证符合标准出厂. Dimensions and surfaces are qualified. No defect in the macrostructure of steels. Hereby certify that all is in accordance with relevant standard.<br>2.有异议时, 来函请标明钢号、炉号、规格、发货单号、原因, 提供质量证明书, 并将实物保管好. If there is any objection, please provide, Steel grade, Heat NO, dimension, Delivery NO., reasons and quality certificate. Please keep the products original.<br>3.建议按生产日期15天后使用 Please use after 15 days of the production date. |                          |             |          |          |            |                 |                        |             |                 |         |    |    |    |                                  |    |    |    |     |    |    |    |     |
| 印章 SEAL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                          |             |          |          |            |                 |                        |             |                 |         |    |    |    |                                  |    |    |    | 吴淑君 |    |    |    |     |